FIF0309

a) $\int \sqrt{1-x^2}\,dx = \int \sqrt{1-\sin^2(\theta)}\cos(\theta)\,d\theta = \int (1-\sin^2(\theta))\cos(\theta)\,d\theta$

$x = \sin(\theta)$

$dx = \cos(\theta)\,d\theta$

$= \int \cos(\theta)\,d\theta - \int \sin(\theta)\cos(\theta)\,d\theta$

$= \int \sqrt{\cos^2(\theta)}\cos(\theta)\,d\theta - \int \cos^2(\theta)\,d\theta$

$= \int \frac{1}{2}\frac{\cos(2\theta)}{2}\,d\theta = \frac{1}{2}\int d\theta + \frac{1}{2}\int \cos(2\theta) = \frac{\theta}{2} + \frac{\sin(2\theta)}{4} = \frac{\arcsin(x)}{2} + \frac{x}{2} + C$



c) $\int \sqrt{x^2-1}\,dx = \int \sqrt{\sec^2(\theta)-1}\sec(\theta)\tan(\theta)\,d\theta = \int \sqrt{\tan^2(\theta)}\sec(\theta)\tan(\theta)\,d\theta$

$x = \sec(\theta)$

$dx = \sec(\theta)\tan(\theta)\,d\theta$

$= \int \tan^2(\theta)\sec(\theta)\,d\theta = \int (-1+\sec^2(\theta))\sec(\theta)\,d\theta = -\int \sec(\theta)\,d\theta + \int \sec^3(\theta)\,d\theta$

$= -\ln|\sec(\theta)+\tan(\theta)| + \int \sec^3(\theta)\,d\theta = \sec(\theta)\tan(\theta) - \frac{\tan^2(\theta)}{2} - \ln|\sec(\theta)+\tan(\theta)| + C$

$\int \sec^2(\theta)\sec(\theta)\,d\theta = \int (\tan^2(\theta)+1)\sec(\theta)\,d\theta = \int \tan^2(\theta)\sec(\theta) + \int \sec(\theta)\,d\theta$

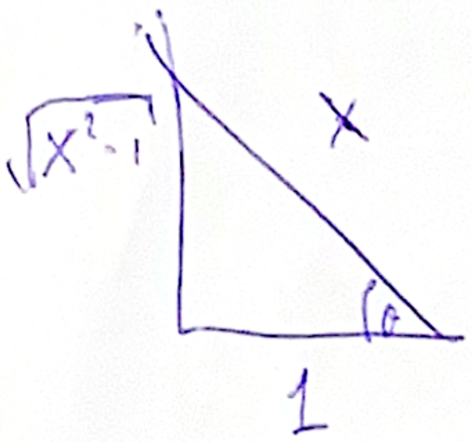
$u = \sec(\theta)$ $dv = \sec^2(\theta)\,d\theta$

$du = \sec\tan(\theta)$ $v = \tan(\theta)$

$\sec(\theta)\tan(\theta) = \int \sec^2(\theta)\tan(\theta)\,d\theta = \sec(\theta)\tan(\theta) - \frac{\tan^2(\theta)}{2}$

$u = \tan(\theta)$ $du = \sec^2(\theta)$

$\int u\,du = \frac{u^2}{2}$



$$\therefore \sec(\theta)\tan(\theta) - \frac{\tan^2(\theta)}{2} - \ln|\sec(\theta)+\tan(\theta)| + C$$

$$= \frac{x\sqrt{x^2-1}}{1} - \frac{x-1}{2} - \ln\left|x+\sqrt{x^2-1}\right| + C$$

e) $\int \sqrt{4+x^2}\,dx$

$$\int \left(16\sec^2(\theta) - 16\right)^{3/2} \cdot 4\sec(\theta)\tan(\theta)\, d\theta$$

$$\int 256\sqrt{(\tan^2(\theta))^3}\,\sec(\theta)\tan(\theta)\, d\theta = 256\int \sec(\theta)\tan^4(\theta)\, d\theta =$$

$$256\int \frac{\sin^4(\theta)}{\cos^4(\theta)} \cdot \frac{1}{\cos(\theta)}\, d\theta = 256\int \frac{\sin^4(\theta)}{\cos^5(\theta)} \cdot \cos(\theta)\, d\theta$$

$$= 256\int \frac{\sin^4(\theta)}{(1-\sin^2(\theta))^3}\cos(\theta)\, d\theta = 256\int \frac{u^4}{(1-u^2)^3} = 256\int \ldots$$

descomposición por fracciones parciales.

$u = \sin(\theta)$

$du = \cos(\theta)$

$$\int \frac{A}{(1-u)^3} + \frac{B}{(1-u)^2} + \frac{C}{(1-u)^3} + \frac{D}{(1+u)} + \frac{E}{(1+u)^2} + \frac{F}{(1+u)^3}\, du$$

a) $\int \frac{2x-5}{x(x+2)}\,dx$

b) $\int \frac{2x+3}{(x-1)(x+2)}\,dx = \int \frac{A}{(x-1)} + \frac{B}{(x+2)}\,dx = \int \frac{\frac{3}{5}}{(x-1)}\,dx + \int \frac{\frac{1}{3}}{(x+2)}\,dx$

$= \frac{3}{5}\ln|x-1| + \frac{1}{3}\ln|x+2|$

$2x+3 = A(x+2) + B(x-1)$

Si $x=-2 \Rightarrow -1 = B(-3) \Rightarrow B = \frac{1}{3}$

Si $x=1 \Rightarrow 5 = A(3) \Rightarrow A = \frac{3}{5}$

d) $\int \frac{3x^2+4x+1}{(x-2)^2(x+1)}\,dx = \int \frac{A}{(x-2)} + \frac{B}{(x-2)^2} + \frac{C}{(x+1)}$